

Data Visualization & Insight Communication



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With assistance from AI tools,
particularly Microsoft Copilot and
Google Gemini

What we cover today

1

Visualization for analytics

Why charts are an analysis tool, not a decoration.

2

Eight analytical purposes

Trend, comparison, ranking, part-to-whole, distribution, relationship, geography, target.

3

Insight communication

Question → Visualization
→ Observation → Insight
→ Implication.

4

Choosing & principles

A selection guide plus twelve rules for honest, readable charts.

Every purpose follows the same teaching pattern: question → chart choice → visualization → interpretation → best practice → mistakes → code.

Why visualization is essential to analysis

Perception is faster than arithmetic

- The eye detects slope, length and clustering in milliseconds.
- A 40-row table of the same numbers takes minutes and still hides the shape.

Summary statistics can lie

- Four datasets can share identical means, variances and correlation yet look completely different.

- Only the plot reveals

Visualization is a thinking tool

Most real insight appears during exploratory plotting.

Visualization does analytical work in four places

Explore

Find shape, gaps and surprises in unfamiliar data.

Verify

Check assumptions: linearity, outliers, skew, balance.

Explain

Make a finding legible to someone who was not in the

Decide

Give a stakeholder enough context to act

What a chart lets you see that a table hides

Patterns

Trends, cycles and seasonality over time — a rising slope is visible instantly.

Comparisons

Which category is larger, and by how much — judged by bar length.

Distributions

Centre, spread, skew and gaps — the shape of a single variable.

Relationships

Whether two variables move together, and how tightly.

Anomalies

Outliers, breaks and data-quality faults that summaries average away.

Rule of thumb: if you cannot name which of these five you are looking for, you are not ready to choose a chart yet. Decide the pattern first, then the geometry.

Showing data ≠ communicating an insight

SHOWING DATA

Title: “Revenue by month”

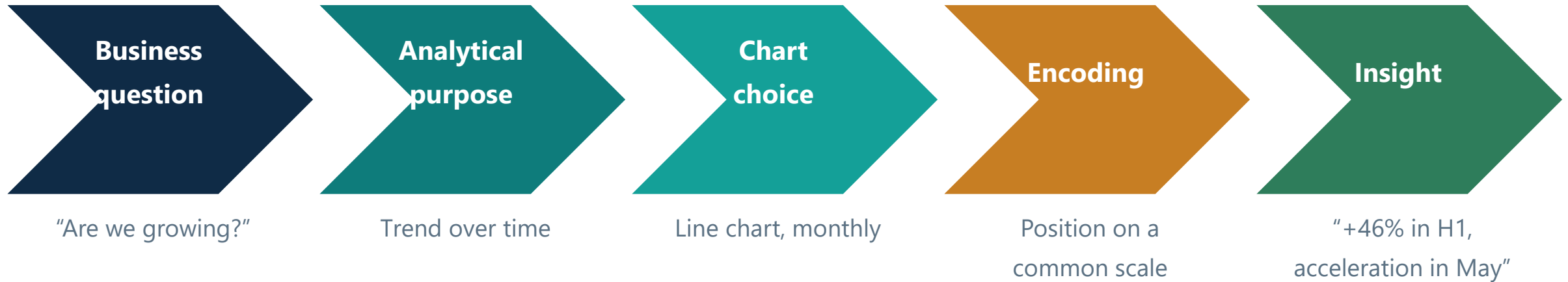
- Describes the axes, not the finding.
- Every series plotted with equal visual weight.
- Reader must do the analysis themselves.
- Caption: “The chart shows monthly revenue from January to June.”
- Correct — and completely uninformative.

COMMUNICATING AN INSIGHT

Title: “Revenue grew 46% in H1, driven by May–June”

- Title states the finding; the chart is the evidence.
- One series highlighted, the rest muted as context.
- Annotation marks the turning point.
- Caption: “Growth accelerated after the May pricing change; Q3 capacity planning should assume the higher baseline.”
- Correct — and actionable.

The analytical question determines the chart



Work forwards, not backwards

Start from the question and let it narrow the options. "I like treemaps" is not an analytical reason. If you cannot write the question in one sentence, the chart will not be clear either.

One chart, one question

A chart that answers three questions usually answers none of them well. Split it, or build a dashboard where each panel owns one question.

02

Eight Analytical Purposes

For each purpose: the question · the chart options · a worked example · how to read it · best practice · common mistakes · Seaborn and Plotly code.

The eight purposes at a glance

1 Trend

How does it change over time?

Line chart

2 Comparison

How do categories differ?

Bar chart

3 Ranking

What is highest or lowest?

Sorted bar

4 Part-to-Whole

How does each part contribute?

Stacked bar

5 Distribution

How are values spread?

Histogram / box

6 Relationship

How are variables related?

Scatter plot

7 Geographic

Where does it happen?

Choropleth map

8 Target / Deviation

How do we compare with plan?

Actual-vs-target bar

Purpose 1 — Trend

ANALYTICAL QUESTION

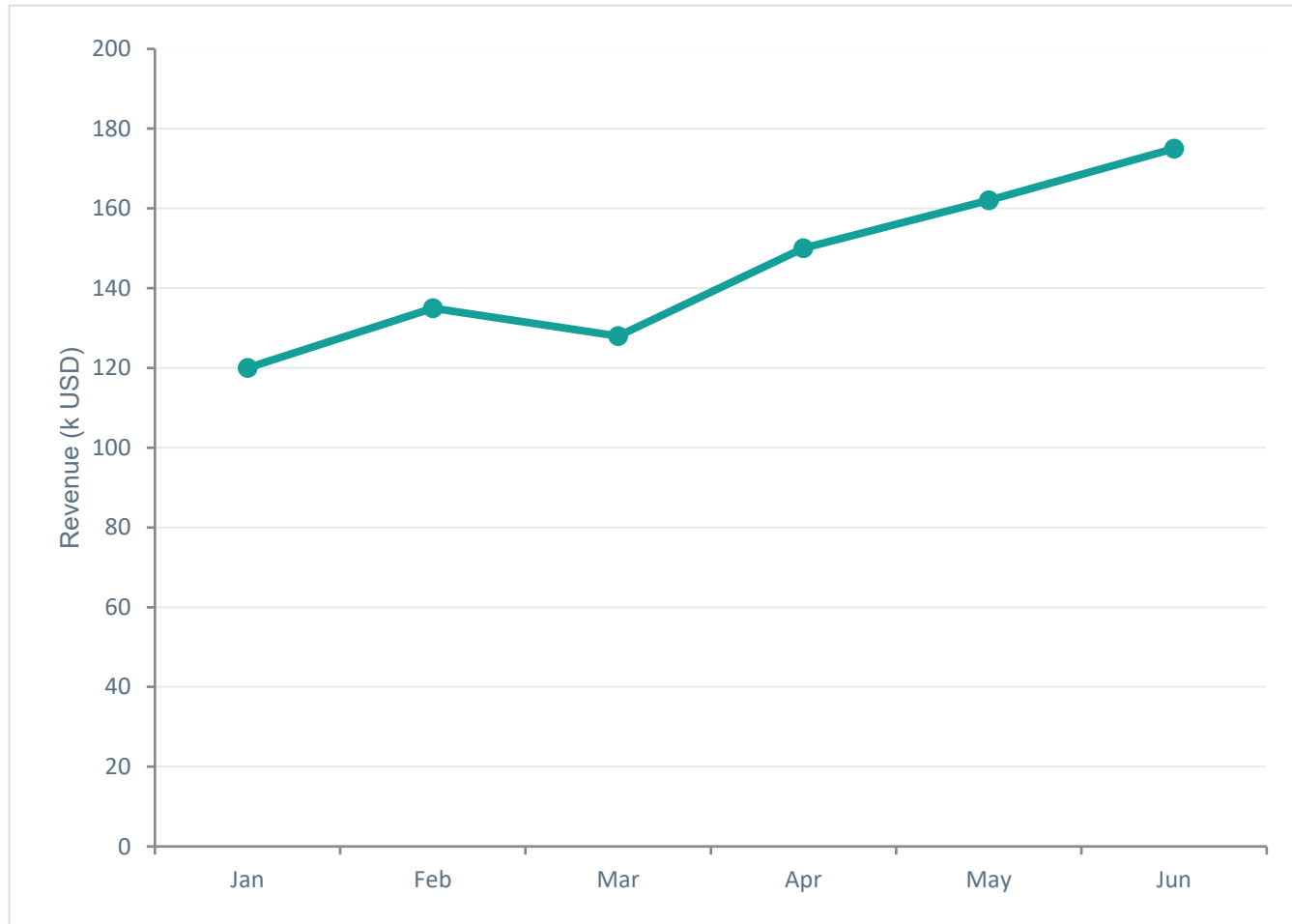
How does the metric change over time?

Recommended charts — and when to use each

★ DEFAULT — LINE CHART	ALTERNATIVE — AREA CHART	ALTERNATIVE — COLUMN CHART
The default. Position over a continuous time axis shows slope, turning points and seasonality clearly. Use for 5+ time points.	Only when the magnitude of a single total matters, or for stacked totals. Avoid overlapping areas for many series.	When time points are few and discrete (4 quarters) and each period is a separate event rather than a flow.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Trend — worked example and interpretation



ShopStream monthly revenue, Jan–Jun

How to read it

Read left to right. Slope is the message: steeper means faster change. Check the y-axis starts at zero before judging how dramatic the rise looks.

What to look for

Direction, rate of change, turning points, seasonality, and any flat or missing segment.

✓ Justified conclusion

“Revenue rose 46% over six months, with the fastest growth between April and June.”

✗ Not justified

“Revenue will reach 250k by December.” Six points cannot support an extrapolation, and no seasonal cycle has been observed yet.

Trend — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Keep time on the x-axis, evenly spaced, oldest on the left.
- Start the value axis at zero for amounts; relax this only for indices and state why.
- Limit to 3–5 lines; highlight one and mute the rest.
- Label the final point directly instead of using a legend.
- State the aggregation level in the title: daily, weekly, monthly.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Truncated y-axis that turns a 3% change into a cliff.
- Uneven time intervals plotted as if evenly spaced.
- Connecting points across a gap where data is missing.
- A partial final period compared with complete ones — it always looks like a crash.
- Spaghetti charts: twelve lines, no hierarchy.

Trend — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'revenue': [120, 135, 128, 150, 162, 175]})

sns.set_theme(style='whitegrid')
ax = sns.lineplot(data=df, x='month', y='revenue',
                  marker='o', linewidth=2.5)
ax.set_ylim(0, 200)
ax.set(xlabel='', ylabel='Revenue (k USD)')
ax.set_title('Revenue grew 46% in H1 2025')
plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'revenue': [120, 135, 128, 150, 162, 175]})

fig = px.line(df, x='month', y='revenue', markers=True,
              title='Revenue grew 46% in H1 2025')
fig.update_yaxes(range=[0, 200],
                  title='Revenue (k USD)')
fig.update_xaxes(title='')
fig.show()
```

Same dataset, same question. Seaborn returns a static Matplotlib axis — ideal for reports and PDFs. Plotly returns an interactive figure with hover values — ideal for dashboards.

Purpose 2 — Comparison

ANALYTICAL QUESTION

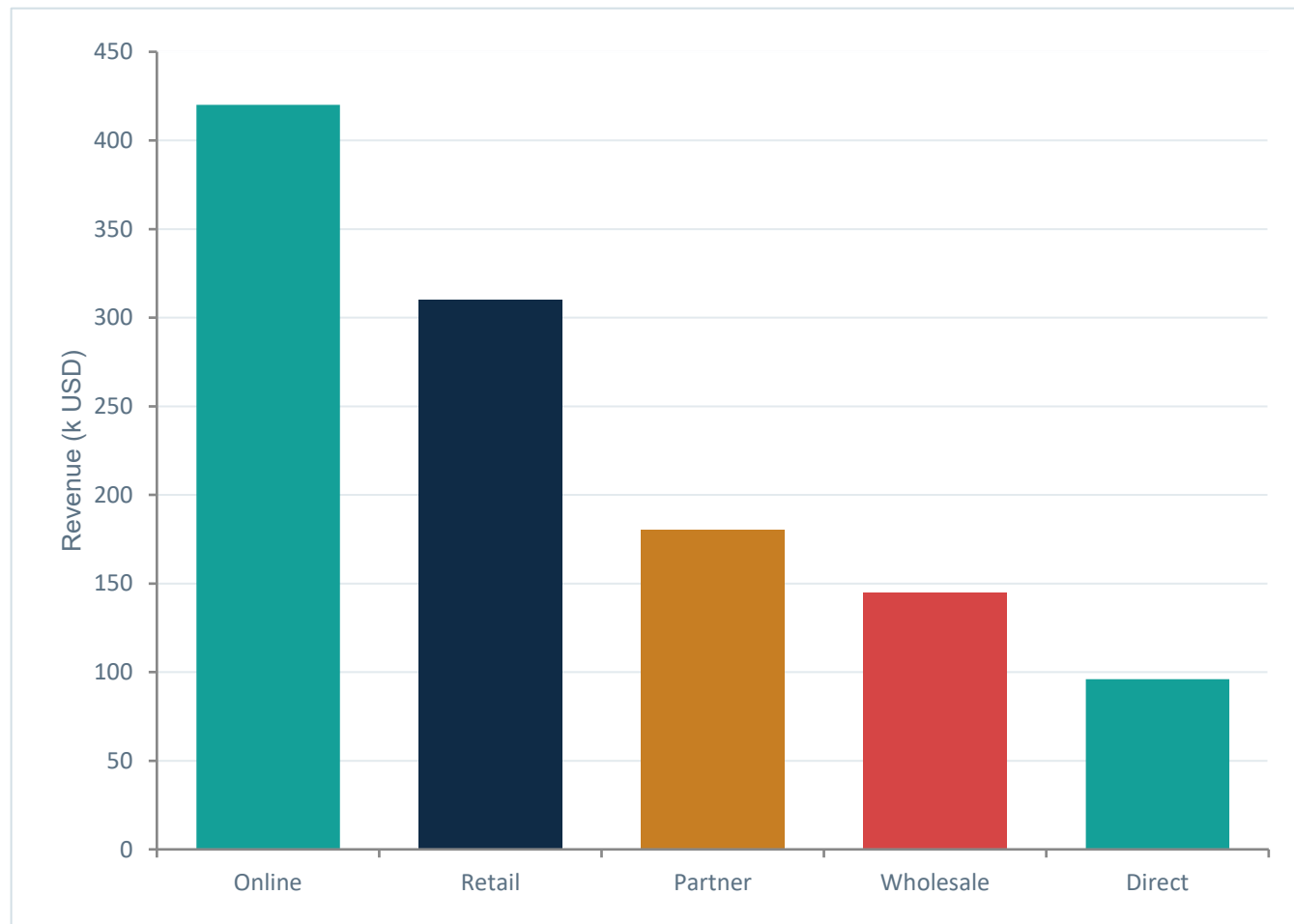
How do categories differ from one another?

Recommended charts — and when to use each

★ DEFAULT — BAR / COLUMN CHART	ALTERNATIVE — GROUPED BAR CHART	ALTERNATIVE — DOT PLOT
The default. Length on a common zero baseline is the most accurately read encoding available.	When each category splits by a second variable (channel × quarter). Keep to 2–3 groups per category.	When you have many categories or long labels — less ink, no baseline crowding.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Comparison — worked example and interpretation



Revenue by sales channel, H1

How to read it

Compare bar lengths against the shared zero baseline. Differences in length are directly proportional to differences in value — that is why the baseline must be zero.

What to look for

Which categories lead, the size of the gap, and whether a few categories dominate the total.

✓ Justified conclusion

"Online generates 4.4× the revenue of Direct and more than a third of H1 revenue across all channels."

✗ Not justified

"Online is our most profitable channel." Revenue is not profit; cost of acquisition is not on this chart.

Comparison — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Always start the value axis at zero — length is the encoding.
- Order bars by value unless the categories have a natural order.
- Use horizontal bars when labels are long.
- One colour for all bars; a second colour only to highlight the point.
- Keep category count under ~12, or switch to a dot plot.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Truncated axis — the classic exaggeration.
- Rainbow colouring that implies a meaning colour does not carry.
- Comparing raw counts across groups of very different sizes instead of rates.
- 3D bars, which make the front row look larger.
- Sorting alphabetically when the question is about magnitude.

Comparison — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'channel': ['Online', 'Retail', 'Partner',
               'Wholesale', 'Direct'],
    'revenue': [420, 310, 180, 145, 96]})

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, x='channel', y='revenue',
                 color='#14A098')
ax.set(xlabel='', ylabel='Revenue (k USD)')
ax.set_title('Online delivers 4.4x the revenue of Direct')
plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'channel': ['Online', 'Retail', 'Partner',
               'Wholesale', 'Direct'],
    'revenue': [420, 310, 180, 145, 96]})

fig = px.bar(df, x='channel', y='revenue',
             text_auto=True,
             title='Online delivers 4.4x the revenue of
Direct')
fig.update_traces(marker_color='#14A098')
fig.update_layout(yaxis_title='Revenue (k USD)',
                  xaxis_title='')

fig.show()
```

Same dataset, same question. Note the single colour in both libraries. Default palettes assign a different colour per bar, which implies a grouping that does not exist. Overriding that is a deliberate honesty choice.

Purpose 3 — Ranking

ANALYTICAL QUESTION

What is highest, what is lowest, and in what order?

Recommended charts — and when to use each

★ DEFAULT — SORTED HORIZONTAL BAR

The default. Sorting does the analytical work; horizontal orientation handles long names.

ALTERNATIVE — TOP-N CHART

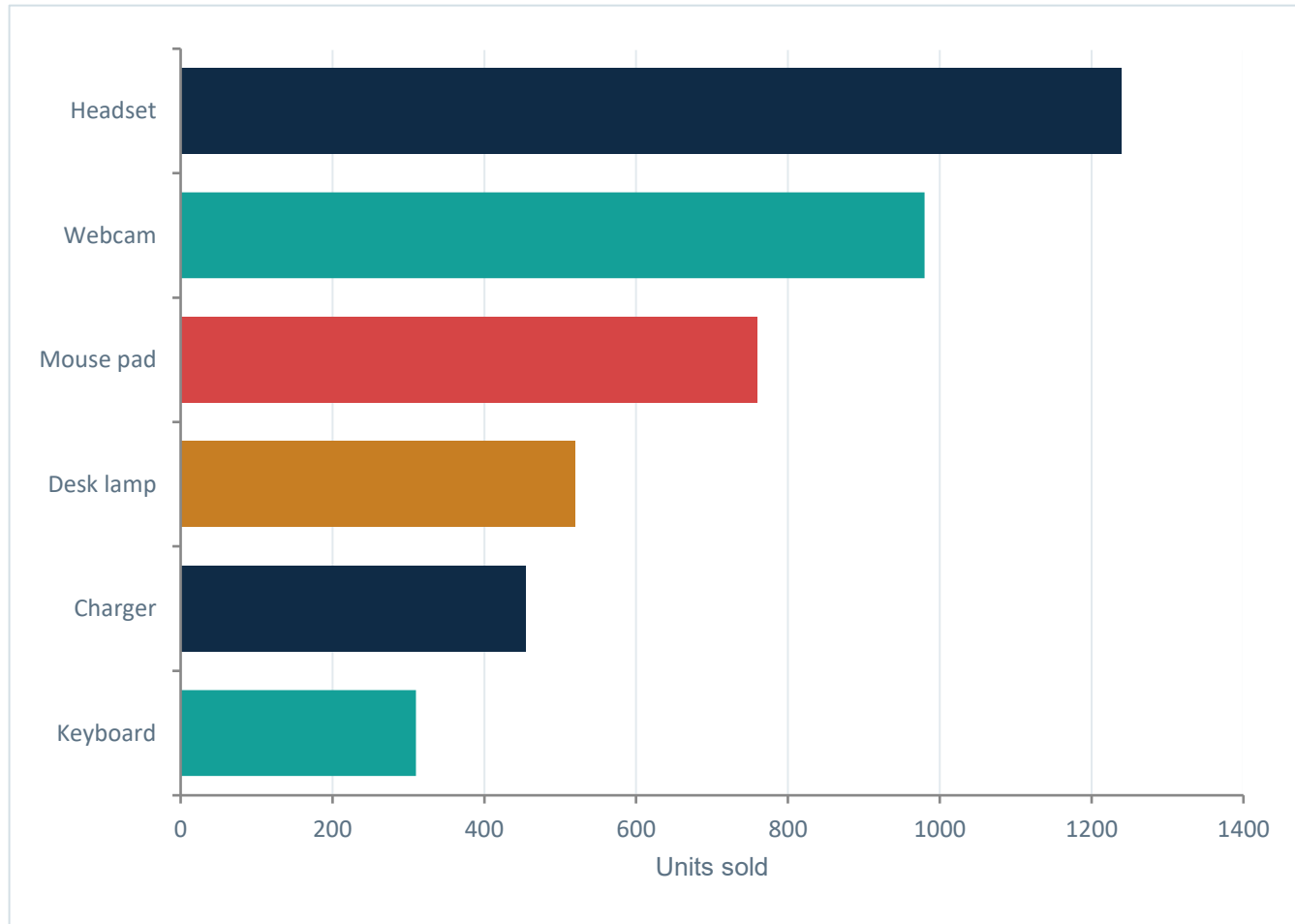
When only the leaders matter.
Always add an 'Other' bar so the reader knows what was excluded.

ALTERNATIVE — LOLLIPOP CHART

Same information, far less ink, when you have 15–30 items.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Ranking — worked example and interpretation



Top six products by units sold, Q2 (sorted)

How to read it

Read from the longest bar down. The order is the message; the lengths tell you whether the gaps between ranks are large or trivial.

What to look for

The leader, the size of the gap to second place, and where the curve flattens into a long tail.

✓ Justified conclusion

"Headsets lead on volume at 1,240 units, 27% ahead of webcams; the top three account for most unit sales."

✗ Not justified

"Headsets are our best product." Units are not revenue or margin — a cheap high-volume item can be the least valuable.

Ranking — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Sort descending; never leave the default alphabetical order.
- Use horizontal bars so names read naturally.
- Show the full ranking or state explicitly how many were cut.
- Highlight only the item the narrative is about.
- Add a value label when exact figures matter to the decision.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Unsorted bars — the reader has to rank them by eye.
- Hidden Top-N: showing five of forty with no 'Other' bar.
- Ranking by a volatile metric on a tiny sample, so the order changes weekly.
- Treating a 2% gap between ranks 1 and 2 as a meaningful difference.
- Ranking on a metric that is not the one the decision depends on.

Ranking — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'product': ['Headset', 'Webcam', 'Mouse pad',
               'Desk lamp', 'Charger', 'Keyboard'],
    'units': [1240, 980, 760, 520, 455, 310]})
df = df.sort_values('units', ascending=False)

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, y='product', x='units',
                 color='#C77E23')
ax.set(ylabel='', xlabel='Units sold')
ax.set_title('Headsets lead Q2 volume, 27% ahead of webcams')
plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'product': ['Headset', 'Webcam', 'Mouse pad',
               'Desk lamp', 'Charger', 'Keyboard'],
    'units': [1240, 980, 760, 520, 455, 310]})

fig = px.bar(df, x='units', y='product', orientation='h',
             title='Headsets lead Q2 volume')
fig.update_yaxes(categoryorder='total ascending')
fig.update_traces(marker_color='#C77E23')
fig.show()
```

Same dataset, same question. Two routes to the same sort: Pandas `sort_values` in Seaborn, versus Plotly's `categoryorder='total ascending'`. Use 'ascending' for horizontal bars because Plotly draws the first category at the bottom.

Purpose 4 — Part-to-Whole

ANALYTICAL QUESTION

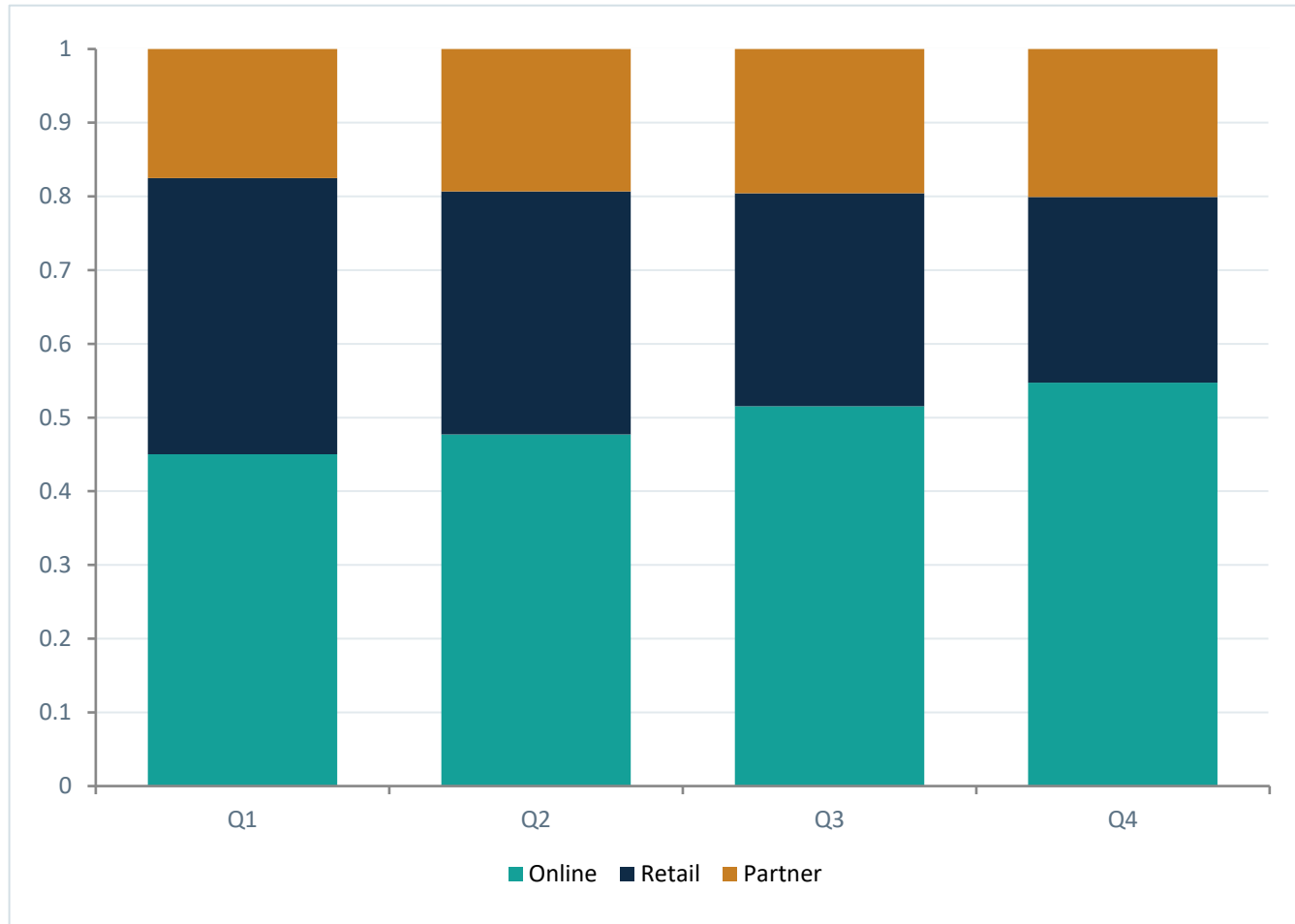
How does each part contribute to the total?

Recommended charts — and when to use each

★ DEFAULT — STACKED BAR CHART	ALTERNATIVE — 100% STACKED BAR	ALTERNATIVE — PIE / DONUT
When absolute totals and composition both matter, especially across several periods.	When only the share mix matters and totals differ wildly between periods.	Only with 2–4 parts and large differences. Angle and area are read poorly; label every slice with its share.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Part-to-Whole — worked example and interpretation



Share of revenue by channel, 100% stacked by quarter

How to read it

Each bar totals 100%. Read one band across the four bars to see whether that share is growing or shrinking. Only the bottom band has a stable baseline.

What to look for

Which share grows, which shrinks, and whether the mix is shifting steadily or abruptly.

✓ Justified conclusion

"Online's share of revenue rose from 45% in Q1 to 55% in Q4, while Retail's share fell by roughly ten points."

✗ Not justified

"Retail is shrinking." Retail's **share** fell; in absolute terms it was broadly flat. Share and amount are different claims.

Part-to-Whole — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Only use part-to-whole when the parts genuinely sum to a meaningful total.
- Put the series you most want tracked at the bottom, on the stable baseline.
- Keep to 3–5 segments; group the rest into 'Other'.
- Label shares directly as percentages.
- Show the absolute total somewhere if the decision depends on size, not just mix.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Pies with ten slices and a legend the reader must decode.
- Stacking categories that overlap or do not sum to the whole.
- Two pies side by side to compare change — use a stacked bar instead.
- Exploded or 3D pies, which distort the area badly.
- Confusing a falling share with a falling amount.

Part-to-Whole — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, matplotlib.pyplot as plt, seaborn as sns

df = pd.DataFrame({
    'quarter': ['Q1', 'Q2', 'Q3', 'Q4'],
    'Online': [180, 210, 250, 300],
    'Retail': [150, 145, 140, 138],
    'Partner': [70, 85, 95, 110]})

share = df.div(df.sum(axis=1), axis=0) * 100
sns.set_theme(style='whitegrid')
ax = share.plot(kind='bar', stacked=True, figsize=(7,4))
ax.set(xlabel='', ylabel='Share of revenue (%)')
ax.set_title("Online's share rose from 45% to 55%")
plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'quarter': ['Q1', 'Q2', 'Q3', 'Q4']*3,
    'channel': ['Online']*4 + ['Retail']*4 + ['Partner']*4,
    'revenue': [180,210,250,300, 150,145,140,138,
                70,85,95,110]})

fig = px.bar(df, x='quarter', y='revenue', color='channel',
             barnorm='percent',
             title="Online's share rose from 45% to 55%")
fig.update_layout(yaxis_title='Share of revenue (%)')
fig.show()
```

Same dataset, same question. Seaborn has no stacked-bar API, so we use Pandas' .plot on a wide frame; Plotly takes long (tidy) format and normalises with barnorm='percent'. Knowing which shape a library wants saves hours.

Purpose 5 — Distribution

ANALYTICAL QUESTION

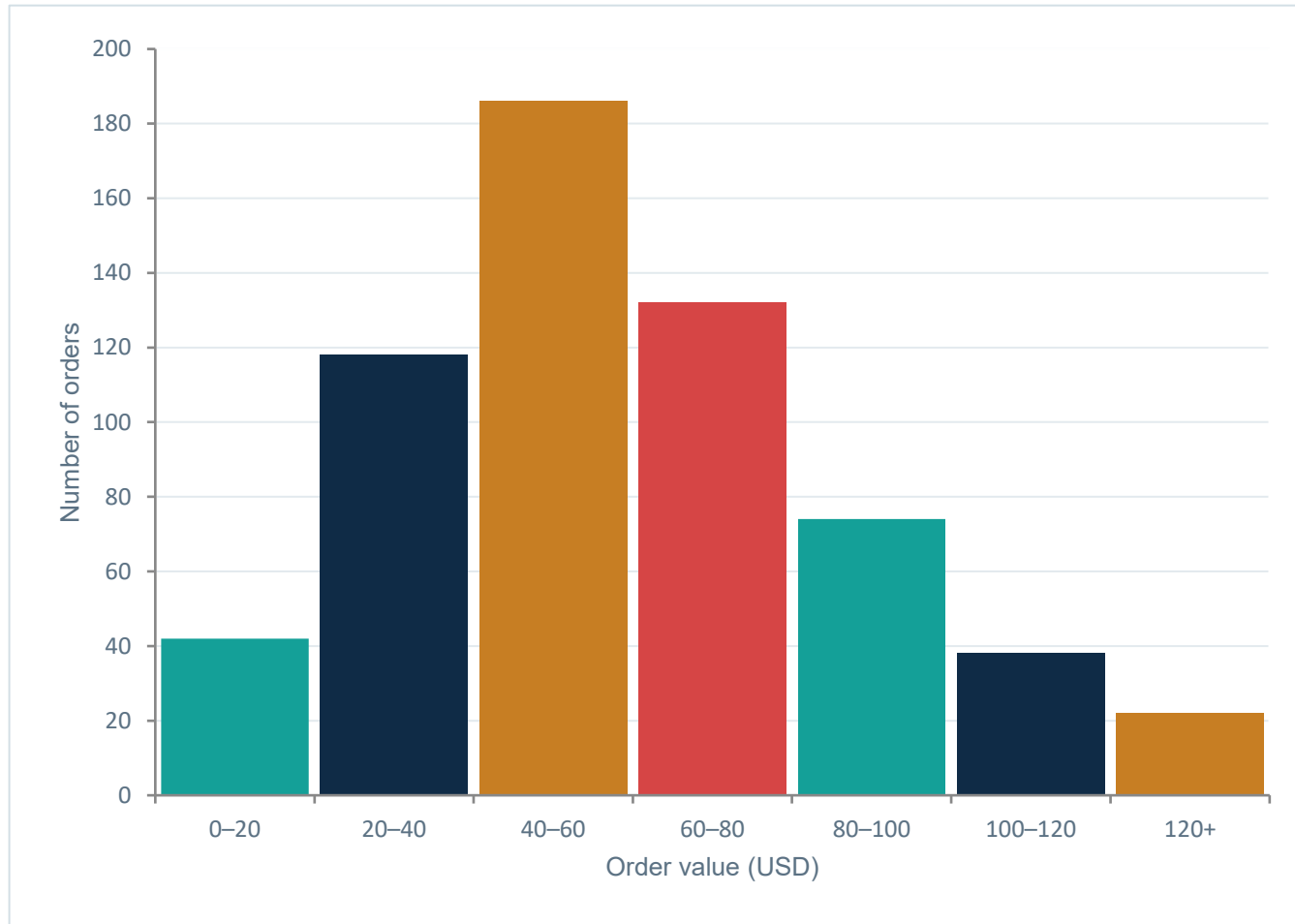
How are the values of one variable spread out?

Recommended charts — and when to use each

★ DEFAULT — HISTOGRAM	ALTERNATIVE — BOX PLOT	ALTERNATIVE — VIOLIN / STRIP PLOT
The default for one variable. Shows centre, spread, skew and gaps. The bin width is an analytical choice.	Best for comparing distributions across groups compactly. Hides multimodality.	When shape matters or the sample is small enough to show individual points.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Distribution — worked example and interpretation



Distribution of order values, 612 orders

How to read it

Each bar counts how many orders fall into that value range. Height is frequency, not amount. The x-axis is a continuous measurement cut into bins.

What to look for

Centre, spread, skew direction, number of peaks, and a long tail or isolated cluster.

✓ Justified conclusion

"Order values are right-skewed: most orders fall between 20 and 80 USD, with a thin tail of high-value orders above 120."

✗ Not justified

"The average order is 60 USD, so a typical customer spends 60." With right skew, the mean is pulled up by the tail — report the median.

Distribution — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Try several bin widths; too few hides structure, too many shows noise.
- Label the axes so frequency is never mistaken for amount.
- Report the median and IQR alongside a skewed distribution.
- Use a box plot when comparing many groups, a histogram for one.
- Say how many observations the distribution is based on.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Bars with gaps, making a histogram look like a bar chart of categories.
- Reporting only a mean for a skewed or bimodal variable.
- Cutting off the tail and hiding the outliers that matter most.
- Using a box plot on a bimodal variable — it looks perfectly normal.
- Comparing histograms with different bin widths or sample sizes.

Distribution — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, numpy as np, seaborn as sns
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
orders = pd.DataFrame({
    'value': rng.gamma(shape=3, scale=18, size=612)})

sns.set_theme(style='whitegrid')
ax = sns.histplot(data=orders, x='value', bins=20,
                  color='#14A098')
ax.axvline(orders['value'].median(), color='#D64545',
          linestyle='--', label='Median')
ax.set(xlabel='Order value (USD)', ylabel='Number of orders')
ax.legend(); plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd, numpy as np
import plotly.express as px

rng = np.random.default_rng(42)
orders = pd.DataFrame({
    'value': rng.gamma(shape=3, scale=18, size=612)})

fig = px.histogram(orders, x='value', nbins=20,
                  marginal='box',
                  title='Order values are right-skewed')
fig.update_layout(xaxis_title='Order value (USD)',
                  yaxis_title='Number of orders')
fig.show()
```

Same dataset, same question. Plotly's `marginal='box'` stacks a box plot above the histogram, so students see both views of the same variable at once — an excellent teaching default.

Purpose 6 — Relationship

ANALYTICAL QUESTION

How are two (or more) variables related?

Recommended charts — and when to use each

★ DEFAULT — SCATTER PLOT

The default for two numeric variables. Shows direction, strength, form and outliers at once.

ALTERNATIVE — BUBBLE CHART

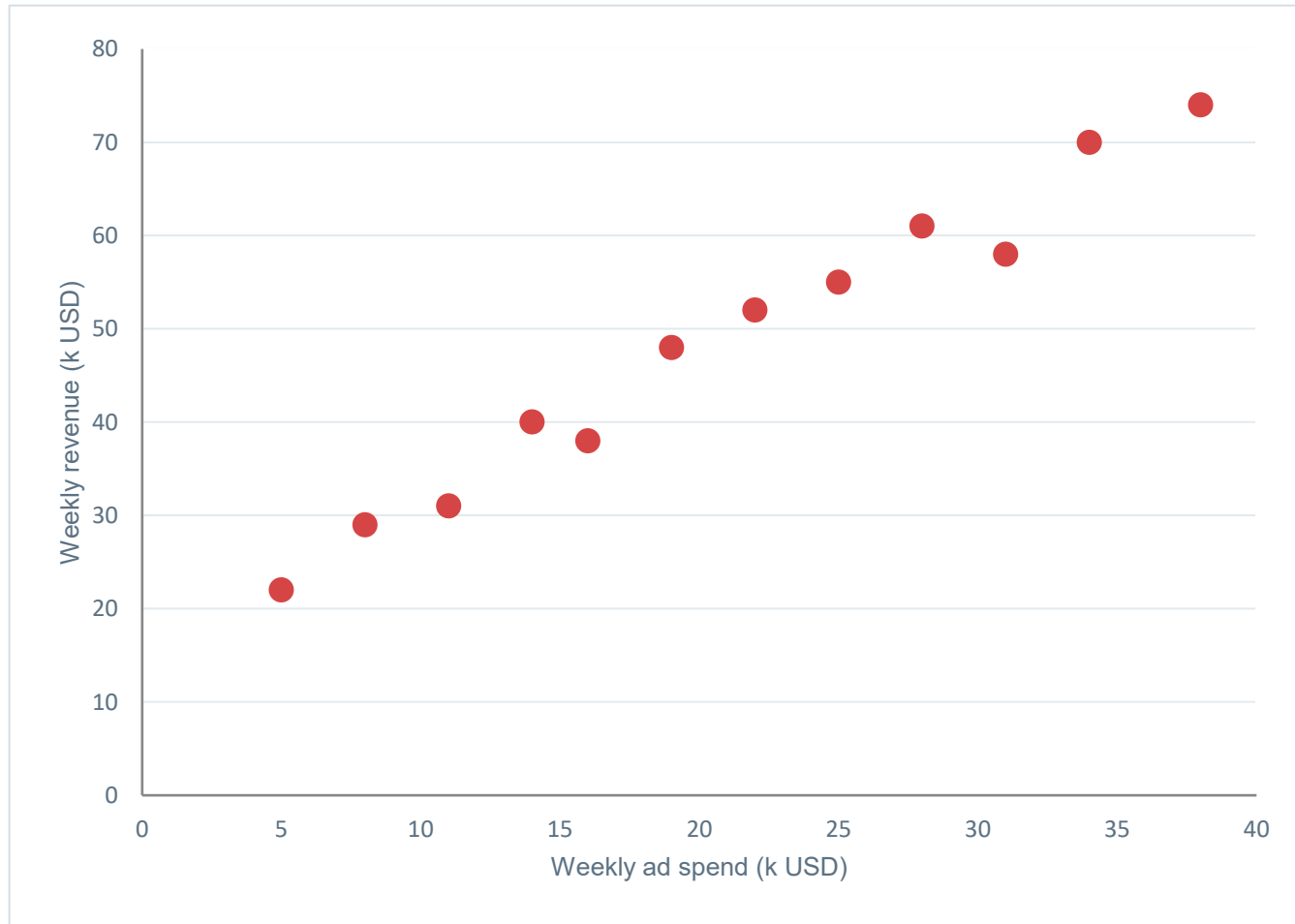
Adds a third variable as point size — use sparingly; area is read inaccurately.

ALTERNATIVE — HEATMAP

For a correlation matrix across many variables, as a screening tool before detailed plots.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Relationship — worked example and interpretation



Weekly ad spend vs revenue, 12 weeks

How to read it

Each point is one week. Horizontal position is spend, vertical position is revenue. Look at the cloud's shape, not at individual points.

What to look for

Direction (positive/negative), strength (tight or loose), form (straight or curved), and points far from the pattern.

✓ Justified conclusion

"Weeks with higher ad spend tend to show higher revenue; the relationship is positive and fairly strong over this range."

✗ Not justified

"Ad spend causes revenue to rise, so doubling spend doubles revenue." Correlation is not causation, and extrapolating beyond 38k is unsupported.

Relationship — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Plot the data before computing a correlation coefficient.
- Put the presumed driver on the x-axis.
- Use transparency or jitter when points overlap heavily.
- Add a trend line only if the form genuinely looks linear.
- State the unit of observation: a week, a customer, a store?

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Claiming causation from a scatter plot alone.
- Reporting r without looking — $r = 0.8$ can describe a curve or an outlier.
- Overplotting that hides thousands of points behind a dense blob.
- Ignoring a lurking variable, such as both rising at Christmas.
- Extrapolating a fitted line beyond the observed range.

Relationship — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'ad_spend': [5,8,11,14,16,19,22,25,28,31,34,38],
    'revenue': [22,29,31,40,38,48,52,55,61,58,70,74]})

sns.set_theme(style='whitegrid')
ax = sns.regplot(data=df, x='ad_spend', y='revenue',
                 color='#D64545', ci=None)
ax.set(xlabel='Weekly ad spend (k USD)',
       ylabel='Weekly revenue (k USD)')
ax.set_title('Higher ad spend is associated with higher revenue')
plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'ad_spend': [5,8,11,14,16,19,22,25,28,31,34,38],
    'revenue': [22,29,31,40,38,48,52,55,61,58,70,74]})

fig = px.scatter(df, x='ad_spend', y='revenue',
                 trendline='ols',
                 title='Ad spend and revenue move together')
fig.update_layout(xaxis_title='Weekly ad spend (k USD)',
                  yaxis_title='Weekly revenue (k USD)')
fig.show() # trendline='ols' needs statsmodels
```

Same dataset, same question. Both add a fitted line in one argument — which is exactly why the wording of the title matters. 'Associated with' is defensible; 'causes' is not.

Purpose 7 — Geographic

ANALYTICAL QUESTION

Where does the pattern happen, and does location explain it?

Recommended charts — and when to use each

★ DEFAULT — CHOROPLETH MAP

Regions shaded by value. Use rates per capita, never raw counts — otherwise you are mapping population.

ALTERNATIVE — SYMBOL / BUBBLE MAP

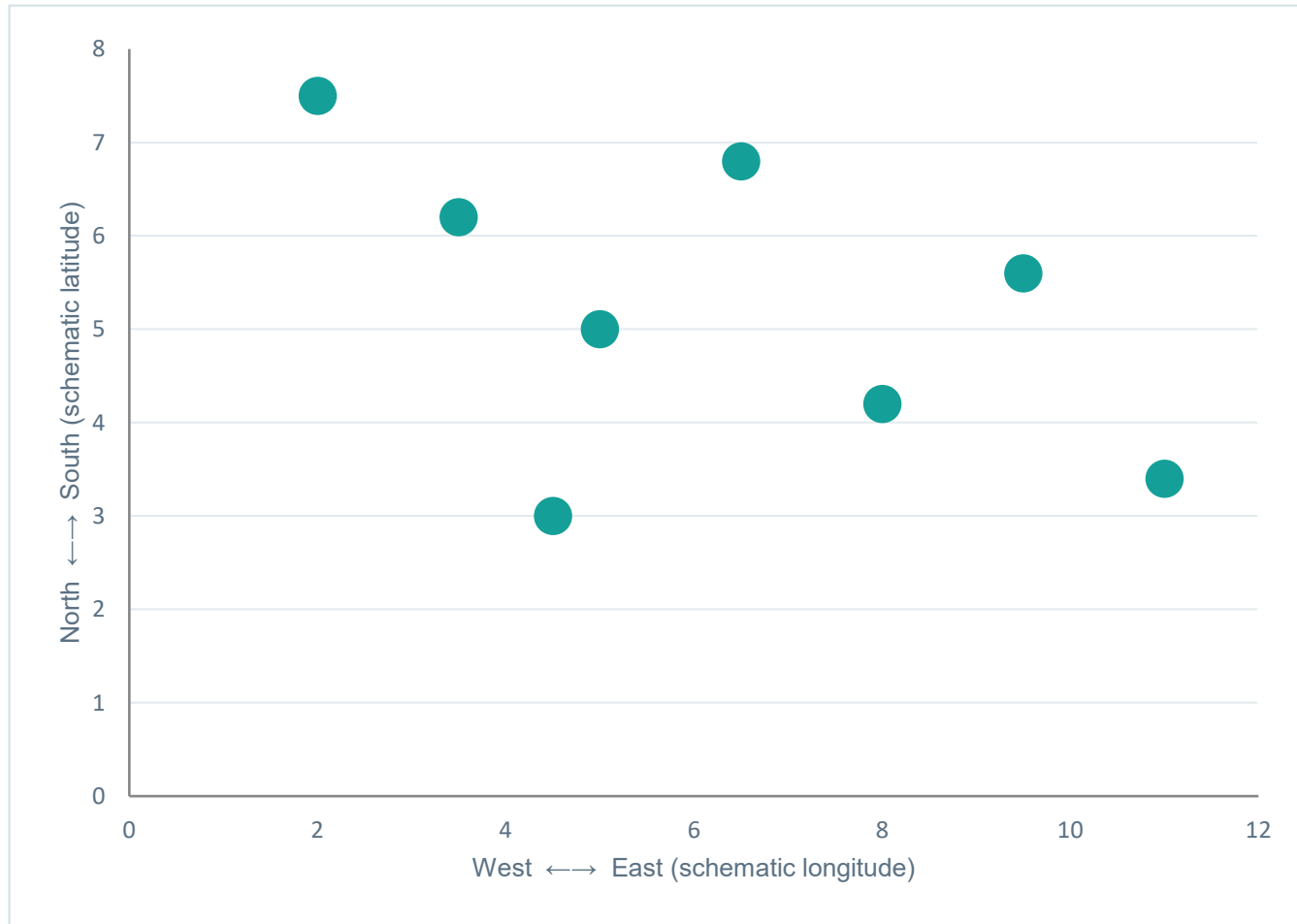
Points sized by value. Better for raw counts and for small areas that a choropleth would hide.

ALTERNATIVE — SORTED BAR BY REGION

Often clearer than a map when you have fewer than ~15 regions and location itself is not the insight.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Geographic — worked example and interpretation



Schematic symbol map: eight city hubs by position

How to read it

Each marker is a city placed at its coordinates. On a real map you would also encode the value as colour or size; position alone only tells you where, not how much.

What to look for

Clusters, isolated hotspots, and whether neighbouring regions behave alike (spatial autocorrelation).

✓ Justified conclusion

"Order rates per 1,000 residents are highest in the northern cluster of hubs, which sit closest to the distribution centre."

✗ Not justified

"The north buys more because northerners prefer our products." Proximity to the warehouse — faster delivery — is a far simpler explanation.

Geographic — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Normalise to a rate: per capita, per household, per 1,000 customers.
- Use a sequential colour scale for magnitude, diverging only around a true midpoint.
- State the geographic unit and the time window.
- Keep map projection and colour bins consistent across comparisons.
- If the map adds nothing a bar chart could not, use the bar chart.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Choropleths of raw counts — a population map in disguise.
- Large rural areas dominating visually while dense cities vanish.
- Arbitrary colour bins chosen to manufacture a pattern.
- Ecological fallacy: inferring individual behaviour from area averages.
- Using a map for decoration when location is irrelevant to the question.

Geographic — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
# Seaborn has no map projection; use lat/lon as a
# schematic scatter, or GeoPandas for real geometry.
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'city': ['Doha', 'Dubai', 'Riyadh', 'Kuwait', 'Muscat'],
    'lon': [51.5, 55.3, 46.7, 47.9, 58.4],
    'lat': [25.3, 25.2, 24.7, 29.4, 23.6],
    'rate': [7.5, 6.2, 5.0, 6.8, 4.2]})

sns.set_theme(style='whitegrid')
ax = sns.scatterplot(data=df, x='lon', y='lat',
                    size='rate', hue='rate', sizes=(60,400),
                    palette='viridis', legend='brief')
ax.set_title('Order rate per 1,000 residents by city')
plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.express as px

df = pd.DataFrame({
    'iso': ['QAT', 'ARE', 'SAU', 'KWT', 'OMN'],
    'rate': [7.5, 6.2, 5.0, 6.8, 4.2]})

fig = px.choropleth(df, locations='iso', color='rate',
                    color_continuous_scale='Viridis',
                    scope='asia',
                    title='Order rate per 1,000 residents')
fig.update_layout(coloraxis_colorbar_title='per 1,000')
fig.show()
```

Same dataset, same question. This is the one purpose where the libraries genuinely differ: Plotly draws real projected maps out of the box, while Seaborn needs GeoPandas. Choose the tool to fit the question.

Purpose 8 — Target / Deviation

ANALYTICAL QUESTION

How does actual performance compare with a target or benchmark?

Recommended charts — and when to use each

★ DEFAULT — ACTUAL-VS-TARGET BAR	ALTERNATIVE — BULLET CHART	ALTERNATIVE — VARIANCE CHART
Bars for actual with a target marker or line. The clearest option for most audiences.	Compact: actual bar, target tick, qualitative bands. Ideal for dashboards with many KPIs.	Plots the gap itself — above and below zero — when the size of the miss is the message.

Pattern: Analytical Question → Chart Choice → Visualization → Interpretation → Insight → Common Mistakes

Target / Deviation — worked example and interpretation



Q4 revenue versus target by region

How to read it

Bars are actual results; the line marks each region's target. The vertical gap between bar top and target marker is the variance — that gap is the entire message.

What to look for

Who is above and below, the size of each gap, and whether misses cluster in one part of the business.

✓ Justified conclusion

"EMEA finished 12% below a 450k target, the only material miss; APAC beat target by 11%."

✗ Not justified

"EMEA underperformed, so the EMEA team is failing." A target may have been set unrealistically, or the market may have moved.

Target / Deviation — best practice and common problems

✓ VISUALIZATION BEST PRACTICE

- Show the target explicitly; a bar alone cannot answer the question.
- Use a consistent colour rule: one colour for above target, one for below.
- Label the variance as both an amount and a percentage.
- State where the target came from and when it was set.
- Order regions by variance so the biggest gap is impossible to miss.

✗ COMMON PROBLEMS AND MISLEADING DESIGNS

- Plotting actuals with no target at all.
- Moving the target mid-period so every result looks like a success.
- Red/green as the only signal — invisible to many colour-blind readers.
- Mixing percentage and absolute variance on one axis.
- Treating a 1% miss and a 30% miss as equally serious.

Target / Deviation — implementation in Seaborn and Plotly

Seaborn · static, report-ready

```
import pandas as pd, seaborn as sns
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'region': ['North', 'South', 'EMEA', 'APAC', 'LATAM'],
    'actual': [520, 480, 396, 610, 340],
    'target': [500, 500, 450, 550, 350]})
df['miss'] = df['actual'] < df['target']

sns.set_theme(style='whitegrid')
ax = sns.barplot(data=df, x='region', y='actual',
                 hue='miss', palette=['#14A098', '#D64545'],
                 dodge=False)
ax.scatter(df['region'], df['target'], marker='_',
          s=900, color='black', label='Target')
ax.set(xlabel='', ylabel='Revenue (k USD)')
ax.legend(); plt.tight_layout(); plt.show()
```

Plotly · interactive, dashboard-ready

```
import pandas as pd
import plotly.graph_objects as go

df = pd.DataFrame({
    'region': ['North', 'South', 'EMEA', 'APAC', 'LATAM'],
    'actual': [520, 480, 396, 610, 340],
    'target': [500, 500, 450, 550, 350]})

fig = go.Figure()
fig.add_bar(x=df.region, y=df.actual, name='Actual',
            marker_color='#14A098')
fig.add_scatter(x=df.region, y=df.target, name='Target',
                mode='markers',
                marker=dict(symbol='line-ew', size=28,
                            line=dict(width=3,
                                        color='black'))))
fig.update_layout(yaxis_title='Revenue (k USD)',
                  title='EMEA is the only material miss (-
12%)')
fig.show()
```

Same dataset, same question. Plotly Express cannot easily mix bars and markers, so we drop to graph_objects. Knowing when to leave the express API is a genuine skill.

03

Insight Communication

A chart is evidence. The analyst's job is the sentence that the evidence supports — and the decision it should change.

A five-step structure that works every time

1. Question	What decision needs an answer?	"Are we growing, and where?"
2. Visualization	Which chart answers it?	Monthly revenue line chart
3. Observation	What does the chart show?	"Revenue rose from 120k to 175k."
4. Insight	What does the pattern mean?	"Growth accelerated after May — the rate changed, not just the level."
5. Implication	Why does it matter, what next?	"Plan Q3 capacity on the higher baseline; verify the pricing change caused it."

Most student answers stop at step 3. The marks — and the professional value — live in steps 4 and 5.

Observation vs insight vs implication

OBSERVATION

What the chart shows

"Online revenue grew 46% between January and June, while Retail was flat."

Descriptive · verifiable from the chart alone · no domain knowledge required.

INSIGHT

What the pattern means

"Growth is coming entirely from one channel, so the business is becoming dependent on Online."

Interpretive · compares the pattern to an expectation · needs context.

IMPLICATION

Why it matters, and what to do

"Concentration risk is rising: protect Online margins and test a second growth channel this quarter."

Decision-oriented · names an action, an owner or a risk.

Test: if your sentence would be identical for any dataset, it is not an insight. If it names no consequence, it is not an implication.

Before → after: turning a description into an insight

Order values

WEAK

"The histogram shows the distribution of order values across the dataset."

STRONG

"Order values are right-skewed: half of all orders are under 48 USD, but a 6% tail above 120 USD drives a quarter of revenue — so free-shipping thresholds should be set from the median, not the mean."

Regional targets

WEAK

"The bar chart compares actual and target revenue for five regions in Q4."

STRONG

"EMEA was the only material miss at 12% below target (-54k); the other four regions met or beat plan, so this is a regional issue, not a company-wide shortfall."

Ad spend

WEAK

"There is a correlation of 0.93 between ad spend and revenue."

STRONG

"Higher-spend weeks are consistently higher-revenue weeks across 5–38k of spend; the association is strong but untested as causal, so a split-region test should run before budget is increased."

Strong statements share three features: a quantified finding, a stated limit on what it proves, and a consequence.

Chart-selection guide: purpose → visualization

Purpose	Analytical question	Recommended	Use instead when...
Trend	How does it change over time?	Line chart, area chart	Few discrete periods → column chart
Comparison	How do categories differ?	Bar chart, grouped bar	Many categories or long labels → dot plot
Ranking	What is highest or lowest?	Sorted horizontal bar	15–30 items → lollipop chart
Part-to-Whole	How does each part contribute?	Stacked bar, 100% stacked bar	2–4 parts, one moment → careful pie/donut
Distribution	How are values spread?	Histogram, box plot	Shape matters or n is small → violin / strip
Relationship	How are variables related?	Scatter plot, heatmap	A third variable matters → bubble chart
Geographic	Where does it happen?	Choropleth (rates), symbol map	<15 regions, location not the point → bar
Target / Deviation	How do we compare with plan?	Bar + target line, bullet chart	The gap is the message → variance chart

Chart selection follows the analytical question — not habit, not the default in the tool, and not personal preference.

Principles 1–6: choose honestly, label clearly

1 · Appropriate chart selection

Let the analytical question choose the geometry. A wrong chart type cannot be rescued by good styling.

2 · Clear titles and labels

The title states the finding; axis labels carry the units. A reader should need no verbal explanation.

3 · Meaningful scales

Linear by default. Use a log scale only for multiplicative data, and say so on the chart.

4 · Honest axes

Zero baseline for bars, always. Never truncate to manufacture drama; never invert to flip a story.

5 · Appropriate aggregation

Daily, weekly or monthly changes the message. Choose the grain the decision needs and state it.

6 · Sorting and ordering

Sort by value unless the categories have a natural order. Order is free analysis.

Principles 7–12: reduce noise, serve the reader

7 • Intentional colour

Colour encodes meaning: highlight, group or scale. If it encodes nothing, use one colour.

8 • No unnecessary 3D

Perspective distorts length, area and angle. 3D adds no data and subtracts accuracy.

9 • Avoid visual clutter

Delete gridlines, borders, shadows and duplicate legends. Maximise the share of ink that is data.

10 • Highlight the finding

One emphasised element, the rest muted. If everything is emphasised, nothing is.

11 • Show enough context

A benchmark, a prior period or a target. A number with no reference point cannot be judged.

12 • Accessibility

Colour-blind-safe palettes, contrast, readable type, and never colour as the only signal.

Six things to carry out of this module

1 The question picks the chart

Name the analytical purpose before you open a plotting library.

2 Length and position beat angle and area

Bars and lines are accurate; pies and 3D are not.

3 Describe, then interpret, then advise

Observation → Insight → Implication, in that order, every time.

4 Say what the data cannot prove

Scoping a claim honestly is what makes it survive challenge.

5 Rates, not counts; median, not mean

When populations differ or data is skewed, the default statistic misleads.

6 The title is where the analysis lands

If the title is a label, the analysis is not finished.